CS-3002 · INFORMATION SECURITY · TOPIC 13

Play fair.
Prove every move.

Cryptographic commitments for rock–paper–scissors tournaments

Pedersen P-256

Commit → Reveal → Verify

THE PROBLEM

The second mover can cheat.

Player A

Reveals rock first

Player B

Chooses paper afterward

A fair game must keep both moves hidden and fixed.

THE PROTOCOL

Commit. Reveal. Verify.

01

Lock

Publish a commitment, not the move.

02

Open

Reveal move and random factor.

03

Check

Verify opening and signature.

PEDERSEN COMMITMENT

Two properties, one point.

$$C = m \cdot g + r \cdot h$$

Hiding

Uniform random r masks the move perfectly.

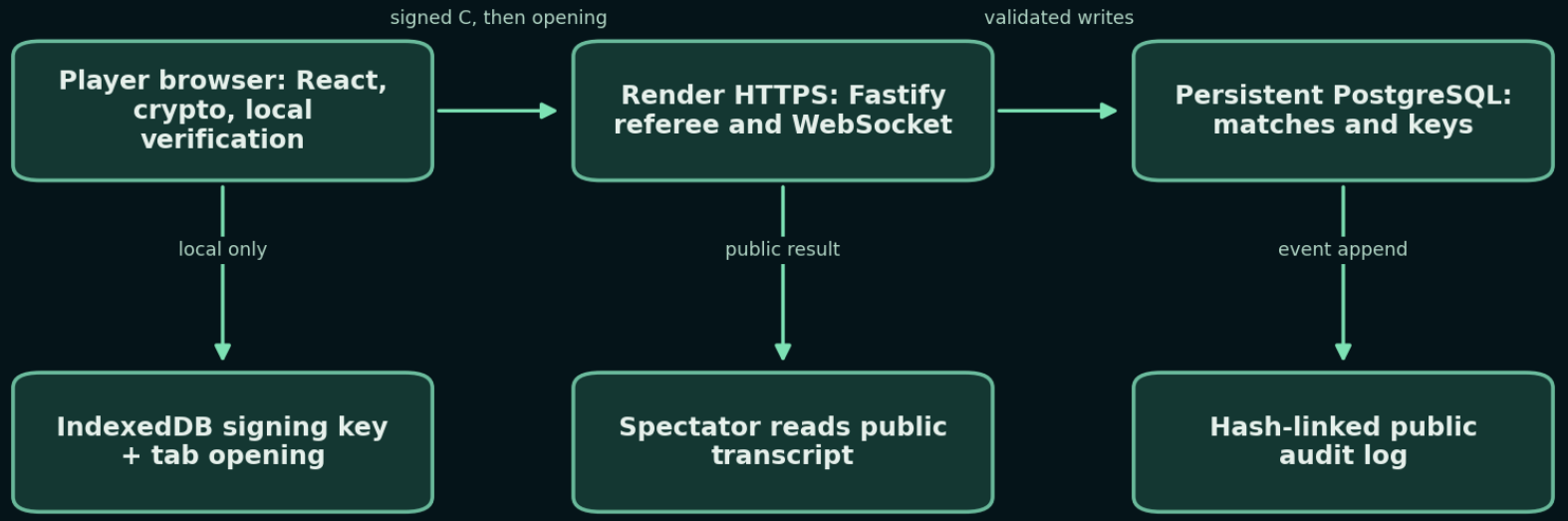
Binding

Changing the move requires a discrete-log trapdoor.

ARCHITECTURE

The browser keeps the secret.

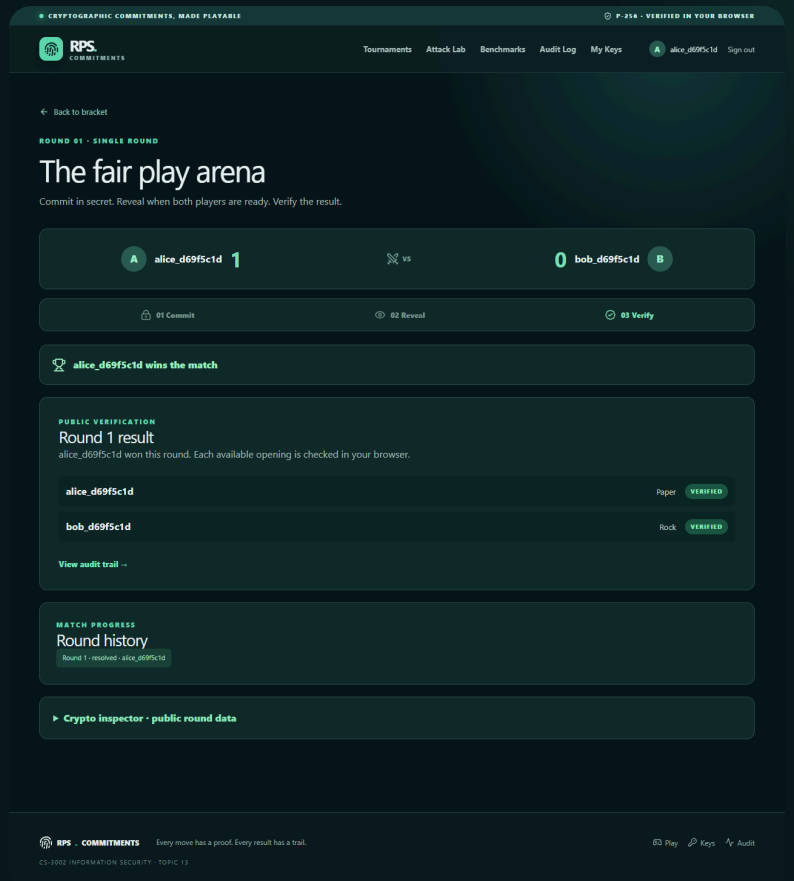
System architecture and trust boundaries



Trust boundary: browser secrets remain local until reveal; referee and database are operator-controlled.

LIVE PROTOCOL

Two browsers. One verifiable result.



ATTACK LAB

Break the weak version.

A1: Unsalted hash

Three guesses reveal the move.

A3: Trapdoor h

One commitment opens two ways.

A9: Replay

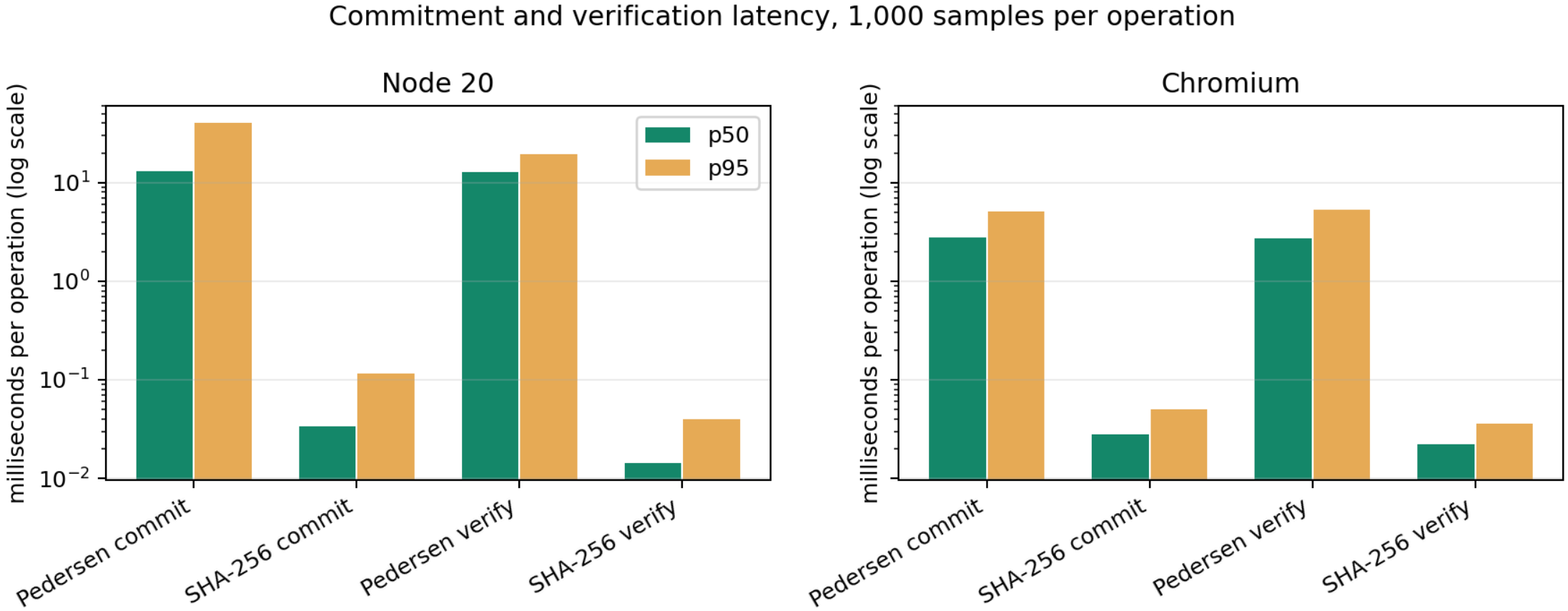
Context-bound signatures stop reuse.

A10: Audit edit

Hash-chain verification detects tampering.

MEASURED PERFORMANCE

Stronger hiding costs CPU.



SECURITY VS USABILITY

One extra byte. More work.

Pedersen

33-byte point · perfect hiding · slower group operations

Salted SHA-256

32-byte digest · computational hiding · faster hashing

Both require a signature and a later opening.

SECURITY BOUNDARIES

What the proof cannot guarantee.

Server

Can leak the first opening or rewrite its whole audit chain.

Device

A compromised browser can expose local secrets.

EVIDENCE AND DEMO

Tested from math to match.

27

unit and API tests

10

attack demonstrations

1,000

samples per benchmark row

Source, data, screenshots, and diagrams are committed to GitHub.

QUESTIONS

Can you trust the evidence?

Openings and signatures: independently checkable. Referee behavior: visible limits.